

Food Healthiness Classification Rubric

DS 4002 – Instructor: Mohammad Ali Khan

Due: TBD

Submission format: Upload link to GitHub repo to Canvas

Individual Assignment

Preparatory Assignments – Consult the reference materials to understand the motivation and the model we will be using.

Why am I doing this? This case study gives the opportunity to explore the data science pipeline through exploring a real-world question in food healthiness classification. You will gain insight into supervised learning with labeling ingredient weight values, understand how a Random Forest Classifier works, and how to validate the model.

What am I going to do? You will build on an existing analysis pipeline that creates rule-based ingredient scores, trains a classification model using those weak labels, and measure the accuracy of created model. Your goal is to understand how the model works, interpret its output, and reflect on the effectiveness of ingredient-based classification.

Tips for success:

- Start by exploring the two csv and understanding their structure and what values they hold
- Test your model with a smaller subset to ensure your code is working before using the whole data set
- When labeling ingredients, explore what type of ingredients tends to affect health the most. Often, unhealthy ingredients have a greater effect than healthy ingredients.
- Make sure to partition each section and comment to keep track of where you are in your code

How will I know I have Succeeded? You will meet expectations on the Food Healthiness Classification project when you follow the criteria in the rubric below.

Spec Category	Spec Details
Formatting	GitHub Repository: It should contain the following items: - The original data set provided in the links and the preprocessed data set all in the same folder - Notebook and/or script files demonstrating preprocessing and analysis - Output files generated during processing and modeling - Executive Summary PDF that explains your findings and conclusions

	- README.md that lays out the project structure, overview, and replication steps
Preprocessing File	Goal: Clean up the data for the health labeling and data analysis step - Join the two provided csv files by products - Filter out any rows with “bad” data - Remove any unneeded columns - Save the clean data in the repository folder
Source Code File	Goal: Implement the labeling logic and build a machine-learning model for predicting healthiness from ingredients. - Define a dictionary or rule-set of ingredient health weights - Tokenize ingredients, score each product based on ingredients, and convert scores into categories - Build a Random Forest Classifier Model using the weak labels - Evaluate Performance with Accuracy, Confusion Matrix, Precision/Recall/F1 - Save all outputs
PDF Report	Goal: Explain your methodology and your findings in a PDF - Summarize the project objective and explain why ingredient-based classification is meaningful - Describe the dataset, preprocessing steps, and any limitations or assumptions - Clearly explain your labeling function design, ingredient weighting strategy, and threshold logic - Describe your model training process, including validation split, hyperparameters, and rationale - Present results with supporting visuals (plots, tables, confusion matrix, evaluation metrics) - Discuss the strengths, weaknesses, and future improvements of your approach
README.md	Goal: Provide a clear guide for anyone trying to understand your project. - Project overview: A short description - Folder structure: Explain where code, data, and outputs are located - Installation instructions: Required libraries, environment setup, and versions - How to run the project: Clear step-by-step instructions - Explanation of outputs: Where to find predictions, charts, and reports

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